

Lab 4 — Code Hint Sheet

The Fashion VAE — a Latent Playground

Only **two** cells need editing — both are creative, so there is no single “right” answer, just a **format** to follow. Type something in and **run the cell**; the built-in **assert** checks confirm the format, and the picture shows the result. Everything else — the model, the training loop, and every plot — runs as-is. The encoder, decoder, and the reparameterization trick are already written for you.

Step 5 — make your own clothes. The VAE’s latent is 2-D, so a “latent point” is just **two numbers**. Every point decodes to something clothes-like — pick a point, see what the VAE draws, then change the numbers and re-run. Roughly $-3 \dots 3$ stays near the clothes the model knows; go further out for stranger results.

Blank 1 — MY_LATENT (a latent point).

- (a) `MY_LATENT = [1.5, -0.8]` — any **list of two finite numbers**. This is the format the cell wants; try `[0, 0]`, `[-2, 1.3]`, `[3, 3]`, ...
- (b) `MY_LATENT = 1.5` — a single number, not a list; trips the check `(len(MY_LATENT) == 2)`.
- (c) `MY_LATENT = [1, 2, 3, 4]` — four numbers; the latent is only 2-D, so this also fails the length check.
- (a) is the format — *any* two finite numbers work, so explore. (b) and (c) are the wrong shape and fail the `len == 2 assert`.

Step 6 — draw anything. Edit the 16×16 ASCII grid: `'#'` is ink, `'.'` is blank. The rule is only about **format** — keep **every row exactly 16 characters** and use only `'#'` and `'.'`. Any shape is fair game; the clothes-VAE will redraw whatever you give it as the nearest garment.

Blank 2 — DOODLE (the drawing).

- (a) Any shape you like — the shipped **spiral**, your initials, a smiley, a star — as long as every row is exactly 16 chars of only `'#'`/`'.'`. This passes and gets reimagined as clothing.
- (b) Rows of *unequal* length, or a stray character like `'x'` or a space — trips the grid sanity-check **assert** (every row must be 16 chars of only `'#'`/`'.'`).
- (c) An empty grid (all `'.'`) — runs fine, but there is nothing to reimagine; draw *something* for a fun rebuild.
- (a) is the goal — there is no wrong drawing, only the format. (b) errors out at the parse check. (c) is legal but boring.

*Answers self-check in the notebook: if the **assert** cells pass — **MY_LATENT** is two finite numbers, and the doodle parses to (1, 784) — you have the format right; the rest is play. The payoff is the tidy VAE latent map (vs. the scattered plain-AE one), the sneaker→boot morph, clothes generated from latent points you choose, and your own doodle snapped to the nearest garment.*